Off on a Tangent

Meditations on the derivative

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Draft of May 21, 2019

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1. Introduction

1.A. Prerequisites

This text assumes familiarity with metric space theory and with linear algebra.

1.B. Linear algebra notation

Definition 1.1. Suppose V is a finite dimensional vector space and $B = (v_1, \ldots, v_n)$ is a basis for V. Then any $v \in V$ can be written uniquely as $a_1 v_1 + \cdots + a_n v_n$ for some scalars $a_1, \ldots, a_n \in \mathbb{R}$, and we define the *representation of v with respect to* B, denoted $[v]_B$, to be the column vector with n entries that records the scalars $a_1, \ldots, a_n$. In other words,

$$[v]_B = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}.$$

Definition 1.2. Suppose $\ell : V \to W$ is a linear map between finite dimensional vector spaces. Let $B = v_1, \ldots, v_m$ be a basis for V and let C be a basis for W. The *representation of ℓ with respect to* B *and* C, denoted $[\ell]_{B,C}$, is the $n \times m$ matrix whose ith column of $[\ell]_{B,C}$ is $[\ell(v_i)]_C$. In other words,

$$[\ell]_{B,C} = \begin{bmatrix} [\ell(v_1)]_C & \cdots & [\ell(v_m)]_C \end{bmatrix}.$$

Lemma 1.3. *If $\ell : V \to W$ is a linear map between finite dimensional vector spaces and B and C are bases for V and W, respectively, then*

$$[\ell]_{B,C}[v]_B = [\ell(v)]_C$$

for any $v \in V$.

Lemma 1.4. *If $\ell : U \to V$ and $\ell' : V \to W$ are linear maps between finite dimensional vector spaces, and A, B and C are bases for U, V, and W, respectively, then*

$$[\ell' \circ \ell]_{A,C} = [\ell']_{B,C}[\ell]_{A,B}.$$

Lemma 1.5. *If $\ell : V \to W$ is a linear map between finite dimensional vector spaces and B and C are bases for V and W, respectively, then the rank of the matrix $[\ell]_{B,C}$ equals the dimension of the range $\ell(V)$.*

Definition 1.6. If $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map, we just write $[\ell]$ to denote the matrix of ℓ with respect to the standard bases on $\mathbb{R}^m$ and $\mathbb{R}^n$.

Lemma 1.7. *If A is a $n \times m$ matrix and $\ell_A : \mathbb{R}^m \to \mathbb{R}^n$ is the linear map $\ell_A(v) = Av$, then*

$$[\ell_A] = A.$$

1.C. Little-oh notation

Throughout this section, let X be a metric space and let x_0 be a point in X.

Definition 1.8. A function $g : X \setminus \{x_0\} \to \mathbb{R}$ is *positive* if $g(x) > 0$ for all $x \in X \setminus \{x_0\}$. Similarly, g is *non-negative* if $g(x) geq 0$ for all $x \in X \setminus \{x_0\}$.

Suppose g is a positive function. Definition 1.9 below formulates what it means for a non-negative function $f : X \setminus \{x_0\} \to \mathbb{R}$ to be "little-oh of g as x tends to x_0." Intuitively, this condition means that f is *much smaller* than g near x_0.

Definition 1.9. A function $f : X \setminus \{x_0\} \to \mathbb{R}$ is *little-oh of* g *as* x *tends to* x_0, written

$$f(x) = o(g(x)) \text{ as } x \to x_0$$

if, for every $\epsilon > 0$, there exists an open neighborhood U of x_0 such that $f(x) \leqslant \epsilon g(x)$ for all $x \in U \setminus \{x_0\}$. Equivalently, this means that

$$\lim_{x \to x_0} \frac{|f(x)|}{g(x)} = 0.$$

When x_0 can be inferred from context, we write simply $f = o(g)$.

Remark 1.10. It's worth noting that the use of the symbol "=" above is mathematically abusive. The left-hand side of the "=" is a function and the right-hand side is a property of functions; of course, a function cannot literally be equal to a property. Rather, the "=" is being used here to mean that the function on the left-hand side has the property described on the right-hand side. As annoying as it is, this abuse of notation is fairly standard, so it's probably best to just get used to it.

Exercise 1.11. Prove that the two conditions in definition 1.9 are in fact equivalent.

Exercise 1.12. Let $X = \mathbb{R}$ and $x_0 = 0$. For each of the functions f and g described below, sketch graphs of f and g and then determine whether or not $f = o(g)$ as $x \to 0$.

(a) $f(x) = x$ and $g(x) = x^2$.

(b) $f(x) = x$ and $g(x) = |x|$.

(c) $f(x) = x^2$ and $g(x) = |x|$.

Exercise 1.13. Let $X = \mathbb{R}^2$ and $x_0 = 0$. For each of the following functions f, determine whether or not $f(x) = o(|x|)$ as $x \to 0$.

(a) $f(x, y) = x - y$.

(b) $f(x, y) = x^2 + y^2$.

Exercise 1.14 ("Smaller than small is still small")**.** Suppose $f_1, f_2 : X \setminus \{x_0\} \to \mathbb{R}$ are functions such that $|f_1(x)| \leqslant |f_2(x)|$ for all x in a punctured neighborhood of x_0, and $f_2(x) = o(g(x))$ as $x \to x_0$. Then $f_1(x) = o(g(x))$ also.

Exercise 1.15. For a fixed positive function $g : X \setminus \{x_0\} \to \mathbb{R}$, prove that the set of functions $f : X \setminus \{x_0\} \to \mathbb{R}$ which are $o(g)$ as $x \to x_0$ is a vector space under the natural operations. In other words, verify the following three facts.

(1) The zero function is $o(g(x))$.

(2) ("Scalar multiples of small are still small") If $c \in \mathbb{R}$ is a scalar and $f(x) = o(g(x))$, then

$$cf(x) = o(g(x)).$$

(3) ("Sum of smalls is still small") If $f_1(x) = o(g(x))$ and $f_2(x) = o(g(x))$, then

$$(f_1 + f_2)(x) = o(g(x)).$$

Exercise 1.16. Suppose $f_1, f_2, g_1, g_2 : X \setminus \{x_0\} \to \mathbb{R}$ are functions with g_1, g_2 positive such that $f_1(x) = o(g_1(x))$ and $f_2(x) = o(g_2(x))$ as $x \to x_0$. Prove that $(f_1 \cdot f_2)(x) = o((g_1 \cdot g_2)(x))$.

1.D. Norms of linear maps

Definition 1.17. Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map. Then we define the *operator norm*, or just *norm*, of ℓ, denoted $\|\ell\|$, by

$$\|\ell\| = \sup\{|\ell(v)| : |v| \leqslant 1\}.$$

Exercise 1.18. Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map. Show that $\ell = 0$ if and only if $\|\ell\| = 0$.

Exercise 1.19. Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map. Show that

$$\|\ell\| = \inf\{\lambda \in \mathbb{R} : |\ell(v)| \leqslant \lambda|v| \text{ for all } v \in \mathbb{R}^m\}.$$

Exercise 1.20 (Submultiplicativity)**.** Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ and $\ell' : \mathbb{R}^n \to \mathbb{R}^p$ are both linear maps. Show that

$$\|\ell' \circ \ell\| \leqslant \|\ell'\| \cdot \|\ell\|.$$

Give an example to show that this inequality can be strict.

Exercise 1.21 ("Linear functions of small functions are still small")**.** Suppose $r : \mathbb{R}^m \to \mathbb{R}^n$ is a function such that $|r(v)| = o(|v|)$ as $v \to 0$. If $\ell : \mathbb{R}^n \to \mathbb{R}^p$ is linear, prove that $|\ell(r(v))| = o(|v|)$ also.

Lemma 1.22. *Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a nonzero linear map. Then there exists $v_0 \in \mathbb{R}^m$ such that $|v_0| = 1$ and $|\ell(v_0)| = \|\ell\|$.*

Proof. The "unit disk"

$$D = \{x \in \mathbb{R}^m : |x| \leqslant 1\}$$

is a nonempty, connected, and compact subset of $\mathbb{R}^m$. The function $v \mapsto |\ell(v)|$ is also a continuous function $D \to \mathbb{R}$, so its image $|\ell(D)|$ must also be a nonempty, connected, and compact subset of $\mathbb{R}$. In other words, it must be a closed interval in $\mathbb{R}$. This means that there exists $v_0 \in D$ such that

$$|\ell(v_0)| = \sup|\ell(D)| = \|\ell\|.$$

Since $\ell \neq 0$, we know $\|\ell\| \neq 0$ from exercise 1.18, which in turn means that $v_0 \neq 0$. To prove that $|v_0| = 1$, assume for a contradiction that $|v_0| < 1$, and let $c = 1/|v_0| > 1$. Then $|cv_0| = |c||v_0| = 1$, so $cv_0 \in D$. But then

$$|\ell(cv_0)| = |c\ell(v_0)| = c|\ell(v_0)| > |\ell(v_0)| = \sup|\ell(D)|,$$

which is absurd. □

Exercise 1.23. Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map and $|\ell(v)| = o(|v|)$ as $v \to 0$. Prove that $\ell = 0$.

2. Single variable derivatives

The goal in this chapter is to formalize derivatives of functions whose input is a single real number and whose output is also a single real number.

2.A. Slope of tangent lines

Let S be a subset of $\mathbb{R}$ and $a \in S$ an interior point. We would like to define the tangent line to the graph of a function $f : S \to \mathbb{R}$ at a. We know intuitively what "tangent line" should mean: a line that "just barely touches" the graph of f at the point $(a, f(a))$. But it's not immediately clear how to formalize this intuition.

One approach to formalization is to notice that our intuitive idea of "tangent line" can be approximated by secant lines, and secant lines can be formalized without ambiguity. More precisely, for small values of h, the point $a + h$ is close to a and still in S since a is an interior point of S. The slope of the secant line passing through the two points $(a, f(a))$ and $(a + h, f(a + h))$ is computed by the quantity

$$\frac{f(a+h) - f(a)}{h}.$$

This quantity is often called a *difference quotient*. See figure 2.1 for a geometric explanation of this difference quotient. By letting $h \to 0$, we get closer and closer to what we intuitively understand to be the slope of the tangent line.

Definition 2.2. A function $f : S \to \mathbb{R}$ is *differentiable* at an interior point $a \in S$ if

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

exists. If this limit does exist, its value is called the *derivative of* f *at* a, and is denoted $f'(a)$.

Exercise 2.3. Use the definition of the derivative to determine whether or not each of the following functions f is differentiable at the given point a. If it is differentiable, calculate $f'(a)$. Check your answers by making sure they agree with geometric intuition and/or rules you learned during your introductory calculus course.

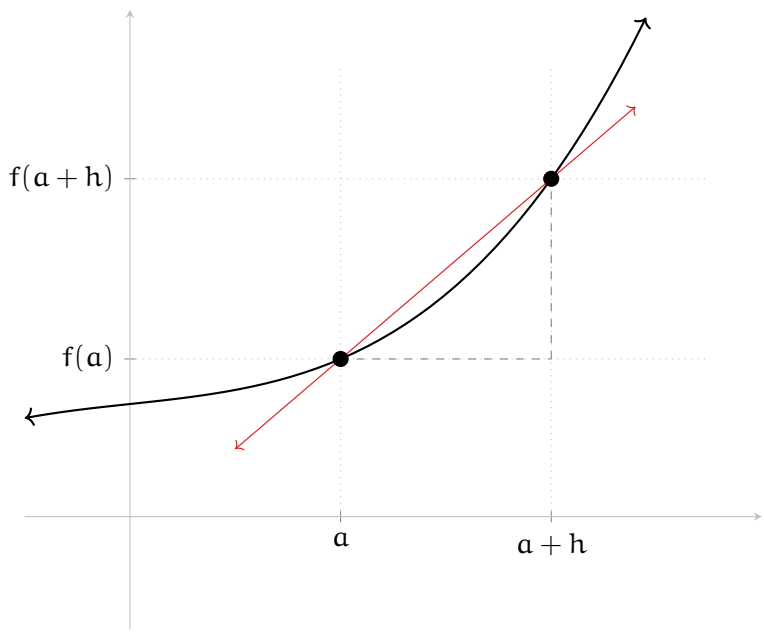

Figure 2.1.: The graph of a function f is depicted in black. The secant line passing through the two points $(a, f(a))$ and $(a+h, f(a+h))$ is depicted in red. The "run" between these points, is h. The "rise" between these points is $f(a+h) - f(a)$. Thus the slope of the secant line, computed as "rise over run," is precisely the difference quotient $(f(a+h) - f(a))/h$.

(a) $f(x) = |x|$ at $a = 0$.

(b) $f(x) = x$ at $a = 0$.

(c) $f(x) = x^2$ at $a = 0$.

(d) $f(x) = x^2$ at $a = 2$.

(e) $f(x) = \begin{cases} 1 & x \geqslant 0 \\ 0 & x < 0 \end{cases}$ at $a = 0$.

Exercise 2.4 (Power rule for positive integer exponents)**.** Prove that, if $f(x) = x^n$ for a positive integer n, then $f'(a) = na^{n-1}$. *Hint.* Recall the binomial theorem, which says that

$$(a+h)^n = \sum_{k=0}^{n} \binom{n}{k} a^k h^{n-k}.$$

Exercise 2.5 (Differentiability implies continuity)**.** Prove that if $f : U \to \mathbb{R}$ is differentiable at $a \in U$, then f is also continuous at a.

2.B. Best linear approximation

Before proceeding further, we reinterpret the definition of the derivative in a way that will be better suited to generalization to the multivariable setting. Suppose a function $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$. The tangent line to the graph at the point $(a, f(a))$ is the graph of the function

$$t(x) = f'(a)(x - a) + f(a).$$

Since this function t is "close" to f near a, we should have that

$$f(a+h) - f(a) \approx t(a+h) - t(a) = f'(a)h$$

for small values of h. See figure 2.6.

Notice that the function $h \mapsto f'(a)h$ is a linear map $\mathbb{R} \to \mathbb{R}$ in the sense of linear algebra. One way to interpret the following result is that the function $h \mapsto f'(a)h$ is the *only* linear map that is a good approximation to $h \mapsto f(a+h) - f(a)$.

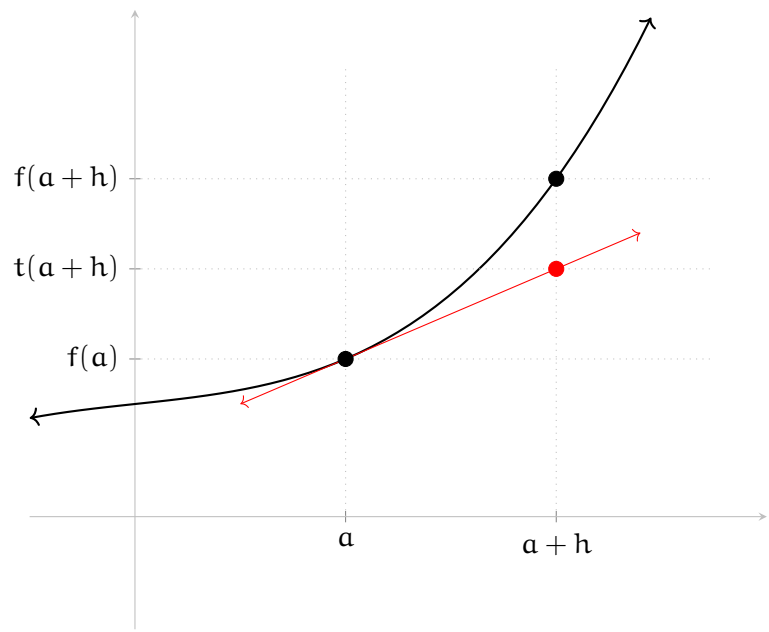

Figure 2.6.: The graph of a function f is depicted in black, and its tangent line t at a is depicted in red. Then the vertical distance $f(a+h) - f(a)$ is approximated by the vertical distance $t(a+h) - t(a)$. Since the tangent line has slope $f'(a)$, this "rise" can be computed as $f'(a)$ times the "run" (in other words, $f'(a)h$). As h gets smaller, this approximation gets better.

Proposition 2.7. *A function* $f : S \to \mathbb{R}$ *is differentiable at an interior point* $a \in S$ *if and only if there exists a linear map* $\ell : \mathbb{R} \to \mathbb{R}$ *such that*

$$|f(a+h) - f(a) - \ell(h)| = o(|h|) \text{ as } h \to 0.$$

Moreover, if there exists such a linear map ℓ*, then* ℓ *is uniquely determined: it must be given by*

$$\ell(h) = f'(a)h$$

for all $h \in \mathbb{R}$*.*

Proof. Let ℓ be a linear map $\ell(h) = ch$, where c is some scalar. Observe that

$$\begin{aligned}\frac{|f(a+h) - f(a) - \ell(h)|}{|h|} &= \left|\frac{f(a+h) - f(a) - ch}{h}\right| \\ &= \left|\frac{f(a+h) - f(a)}{h} - c\right|.\end{aligned}$$

This leads to the following sequence of "if and only if" statements.

$$\begin{aligned}|f(a+h) - f(a) - \ell(h)| = o(|h|) &\iff \lim_{h\to 0} \frac{|f(a+h) - f(a) - \ell(h)|}{|h|} = 0 \\ &\iff \lim_{h\to 0} \left|\frac{f(a+h) - f(a)}{h} - c\right| = 0 \\ &\iff \lim_{h\to 0} \left(\frac{f(a+h) - f(a)}{h} - c\right) = 0 \\ &\iff \lim_{h\to 0} \frac{f(a+h) - f(a)}{h} = c\end{aligned}$$

The statement of the proposition follows. □

Definition 2.8. If $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in U$, we define $df_a : \mathbb{R} \to \mathbb{R}$ to be the linear map $df_a(h) = f'(a)h$.

In other words, df_a is the only good linear approximation to the function $h \mapsto f(a+h) - f(a)$. Being the only good linear approximation, it is also the best linear approximation, whence the name of this section.

2.C. Rules of differentiation

The goal in this section is to prove the rules for differentiation that one encounters in an introductory calculus course.

2.C.1. Sum and scalar multiples rule

Both of the following can be proved using either the definition of the derivative 2.2 or the equivalent "best linear approximation" characterization of proposition 2.7. My suggestion is that you try both methods for both of these exercises. When you use proposition 2.7, you may find exercise 1.15 useful.

Exercise 2.9 (Sum rule)**.** Prove that, if $f, g : S \to \mathbb{R}$ are both differentiable at an interior point $a \in U$, then $f + g$ is also differentiable at a and

$$(f + g)'(a) = f'(a) + g'(a).$$

Exercise 2.10 (Scalar multiples rule)**.** Prove that, if c is a constant and $f : S \to \mathbb{R}$ is differentiable at an interior point $a \in S$, then cf is also differentiable at a and

$$(cf)'(a) = cf'(a).$$

2.C.2. Product and quotient rules

Proposition 2.11 (Product rule)**.** *Suppose* $f, g : U \to \mathbb{R}$ *are both differentiable at* $a \in U$*. Then* fg *is also differentiable at* a *and*

$$(fg)'(a) = f'(a)g(a) + f(a)g'(a).$$

Proof. The proof is a clever algebraic manipulation of difference quotients. The key trick is introducing a "cross term" of the form $f(a)g(a + h)$, and then cancelling it out by also introducing its negative. More precisely, for any sufficiently small real number h, we have

$$\begin{aligned}
\frac{(fg)(a+h) - (fg)(a)}{h} &= \frac{f(a+h)g(a+h) - f(a)g(a)}{h} \\
&= \frac{f(a+h)g(a+h) - f(a)g(a+h) + f(a)g(a+h) - f(a)g(a)}{h} \\
&= \frac{f(a+h)g(a+h) - f(a)g(a+h)}{h} + \frac{f(a)g(a+h) - f(a)g(a)}{h} \\
&= \frac{f(a+h) - f(a)}{h} \cdot g(a+h) + f(a) \cdot \frac{g(a+h) - g(a)}{h}.
\end{aligned}$$

Taking the limit as $h \to 0$ yields the result. □

Exercise 2.12. Check your understanding of the above proof of the product rule.

(a) Could you have instead introduced a cross term of the form $f(a + h)g(a)$?

(b) At what point is exercise 2.5 tacitly used in the above proof?

Exercise 2.13. Reprove the scalar multiples rule (exercise 2.10) using the product rule.

Exercise 2.14 (Quotient rule)**.** Suppose $f, g : S \to \mathbb{R}$ are both differentiable at an interior point $a \in U$, and that $g(a) \neq 0$. Prove that f/g is also differentiable at a and

$$(f/g)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{g(a)^2}.$$

Hint. Observe that

$$\frac{(f/g)(a+h) - (f/g)(a)}{h} = \frac{\frac{f(a+h)}{g(a+h)} - \frac{f(a)}{g(a)}}{h}.$$

Multiply both the numerator and denominator by $g(a+h)g(a)$ to clear the denominators inside the numerator, and then introduce the cross term $f(a)g(a)$. Be sure to notice at what point you use exercise 2.5.

Exercise 2.15 (Power rule for integer exponents)**.** Extend the result of exercise 2.4 to arbitrary integer exponents. *Hint.* Use the quotient rule for negative integers. The exponent zero is a special case; deal with it separately.

2.C.3. Chain rule

Proposition 2.16 (Chain rule)**.** *Suppose that S and T are both subsets of $\mathbb{R}$, that $f : S \to T$ is differentiable at an interior point a, that $f(a)$ is an interior point of T, and that $g : T \to \mathbb{R}$ is differentiable at $f(a)$. Then the composite $g \circ f : S \to \mathbb{R}$ is also differentiable at a, and*

$$(g \circ f)'(a) = g'(f(a))f'(a).$$

Let us discuss two different proofs of this important result.

First proof

The first proof uses the definition of the derivative 2.2, but there is a subtle technicality involved; so, let us discuss it informally first before diving into the formal proof.

The idea involved is again a clever rewriting of difference quotients. Again, we introduce a cross term, which in this is $f(a+h) - f(a)$. But this time we introduce this cross term "multiplicatively" rather than "additively." More precsiely, we have

$$\begin{aligned}\frac{(g \circ f)(a+h) - (g \circ f)(a)}{h} &= \frac{g(f(a+h)) - g(f(a))}{h} \\ &= \frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} \cdot \frac{f(a+h) - f(a)}{h}.\end{aligned}$$

The difference quotient on the right is precisely $f'(a)$. Thus, it would be sufficient to prove

$$\lim_{h\to 0} \frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} = g'(f(a)). \tag{2.17}$$

This formula looks an awful lot like the difference quotient that is used to define the derivative of g at $f(a)$. In fact, since f is continuous at a by exercise 2.5, we have $f(a+h) \approx f(a) + h$ for small values of h, which means that

$$\frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} \approx \frac{g(f(a)+h) - g(f(a))}{f(a)+h-f(a)} = \frac{(g\circ f)(a+h) - (g\circ f)(a)}{h}$$

for small values of h. Taking the limit as $h \to 0$ on the right-hand side yields precisely $g'(f(a))$, so equation (2.17) seems like a reasonable conjecture.

The problem is that equation (2.17) isn't true. Issues arise when $f(a+h) = f(a)$ for arbitrarily small values of h, which leads to an infinite sequence of divisions by 0.

Example 2.18. Let

$$f(x) = \begin{cases} x^2 \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

and let $g(x) = x^2$. Observe that $g'(f(0)) = 0$. You will show later (in exercise 2.29) that f is in fact differentiable at 0, so these functions do satisfy the hypothesis of the chain rule. However, the limit as $h \to 0$ of

$$\frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)} = \frac{(h^2 \sin(1/h))^2}{h^2 \sin(1/h)}$$

does not exist. For example, the sequence $h_k = 1/k\pi$ converges to 0 as $k \to \infty$, but the expression above is undefined along this sequence.

We will get around the kind of issue that is described by the above example by carefully studying the expression

$$\frac{g(f(a+h)) - g(f(a))}{f(a+h) - f(a)}. \tag{2.19}$$

Observe that, when $f(a+h) \neq f(a)$, then this expression is precisely the difference quotient that calculates the slope of the secant line passing through $(f(a), g(f(a))$ and $(f(a+h), g(f(a+h))$. When $f(a+h) = f(a)$, this secant line doesn't make sense; but what does make sense is the tangent line! So we'll remove the undefinedness of expression (2.19) by extending it so that it takes the value $g'(f(a))$ whenever $f(a+h) = f(a)$.

This leads us to our first proof of the chain rule.

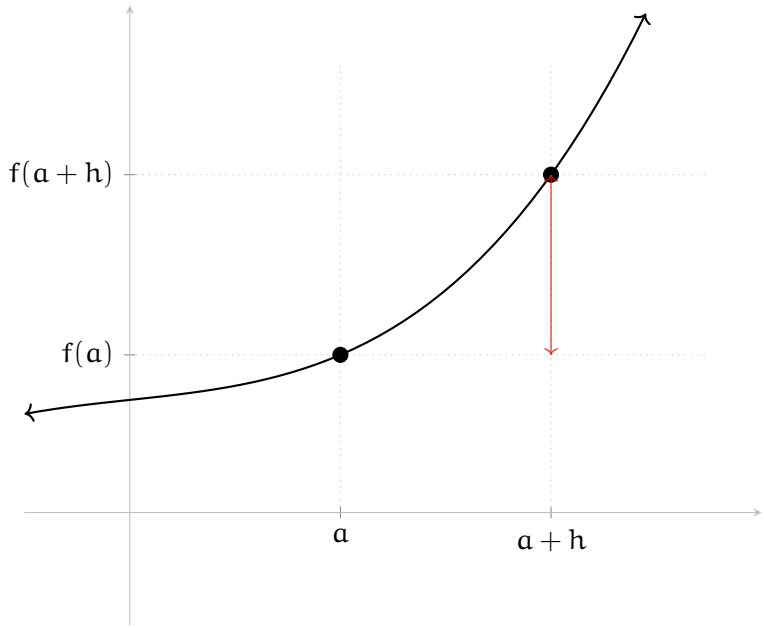

Figure 2.20.: The graph of f is depicted in black. The value of the function t on a small nonzero input value h is the vertical distance depicted in red.

First proof of the chain rule. Consider the function ϵ defined for small values of h by $\epsilon(h) = f(a+h) - f(a)$. See figure 2.20. You will verify in exercise 2.23 below that ϵ is continuous at 0.

Next, define a function σ by the following.

$$\sigma(k) = \begin{cases} \dfrac{g(f(a)+k) - g(f(a))}{k} & \text{if } k \neq 0 \\ g'(f(a)) & \text{if } k = 0 \end{cases}$$

In other words, σ is the function that, on some small nonzero input k, outputs the slope of the secant line connecting the two points on the graph of g where $x = f(a)$ and $x = f(a) + k$. See figure 2.21. It follows from the definition of the derivative of g at $f(a)$ that the function σ is continuous at 0.

Observe that

$$\frac{(g \circ f)(a+h) - (g \circ f)(a)}{h} = \sigma(\epsilon(h)) \cdot \frac{f(a+h) - f(a)}{h} \tag{2.22}$$

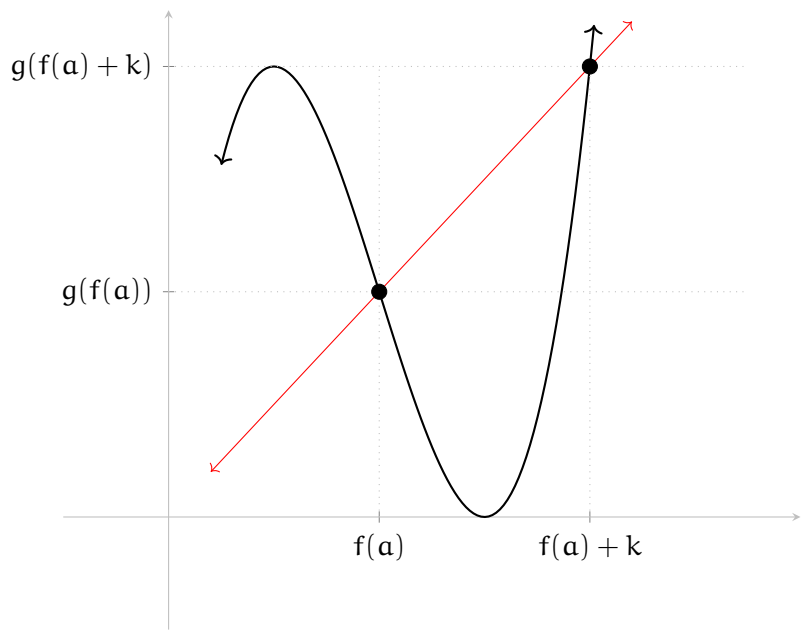

Figure 2.21.: The graph of g is depicted in black. The output of the function s on a small nonzero input value h is the slope of the red secant line passing through the two points $(f(a), g(f(a)))$ and $(f(a)+k, g(f(a)+k))$.

for all small values of h, as you will verify in exercise 2.23 below. Thus

$$\begin{aligned}\lim_{h\to 0}\frac{(g\circ f)(a+h)-(g\circ f)(a)}{h} &= \lim_{h\to 0}\sigma(\epsilon(h))\cdot\frac{f(a+h)-f(a)}{h}\\ &\overset{(*)}{=} \sigma(\epsilon(0))f'(a)\\ &= g'(f(a))f'(a)\end{aligned}$$

where we have used the continuity of ϵ and σ and the definition of the derivative of f at a for the step labeled $(*)$ above. This proves the chain rule. □

Exercise 2.23. Verify the missing details in the above proof. Specifically, check the following.

(a) Prove that ϵ is continuous at 0.

(b) Prove equation (2.22). *Hint.* As suggested by our analysis before the proof of the chain rule, split up your proof of equation (2.22) into two cases: when $f(a+h)\neq f(a)$, and when $f(a+h)=f(a)$.

Second proof

Our second proof of the chain rule will use the "best linear approximation" interpretation of derivatives provided by proposition 2.7.

Second proof of the chain rule. By proposition 2.7, it is sufficient to prove that

$$g(f(a+h))-g(f(a))-dg_{f(a)}(df_a(h))=o(|h|) \text{ as } h\to 0.$$

Define the following "error in approximation" functions.

$$\begin{aligned}r(h) &= f(a+h)-f(a)-df_a(h)\\ s(k) &= g(f(a)+k)-g(f(a))-dg_{f(a)}(k)\end{aligned}$$

See figure 2.24.

Since

$$f(a+h)=f(a)+df_a(h)+r(h)$$

by definition of r, we have

$$\begin{aligned}g(f(a+h)) &= g(f(a)+df_a(h)+r(h))\\ &= g(f(a))+dg_{f(a)}(df_a(h)+r(h))+s(df_a(h)+r(h))\\ &= g(f(a))+dg_{f(a)}(df_a(h))+dg_{f(a)}(r(h))+s(df_a(h)+r(h))\end{aligned}$$

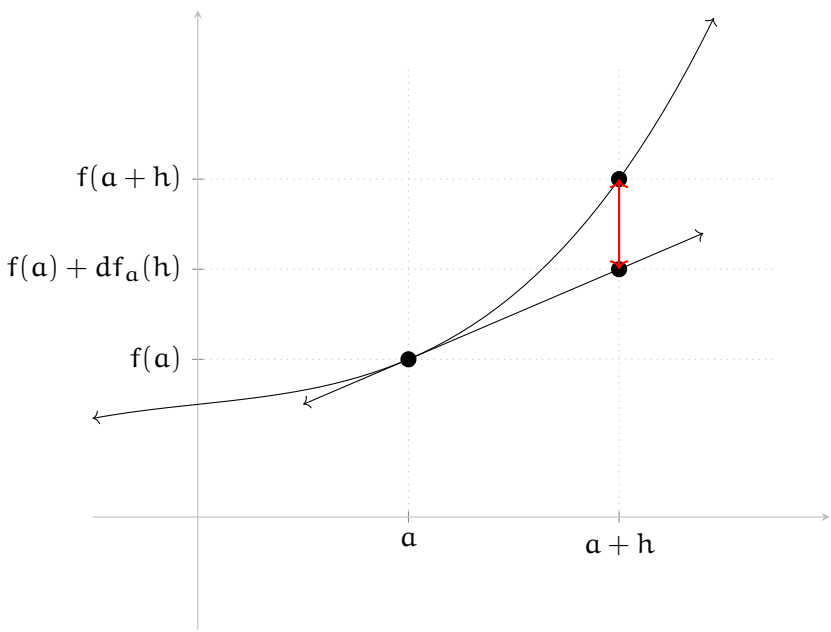

Figure 2.24.: The graphs of a function f and its tangent line $f(a) + df_a$ are depicted in red. On a small input h, the "error in approximation" function r outputs the vertical distance indicated in red.

where we use the definition of s on input $k = df_a(h) + r(h)$ for the second step, and the fact that $dg_{f(a)}$ is linear for the third. Thus

$$g(f(a+h)) - g(f(a)) - dg_{f(a)}(df_a(h)) = dg_{f(a)}(r(h)) + s(df_a(h) + r(h)).$$

We therefore want to prove that the right-hand side is $o(|h|)$.

Since

$$dg_{f(a)}(r(h)) = g'(f(a)) \cdot r(h)$$

is just a scalar multiple of r, and $r = o(|h|)$, we see that $dg_{f(a)}(r) = o(|h|)$ also. So it is sufficient to prove that

$$s(df_a(h) + r(h)) = o(|h|).$$

We've now hit the technical part, so let's hit pause.

To see why this is technical, and for essentially the same reasons as the previous proof, notice that we're trying to prove that

$$\lim_{h \to 0} \frac{|s(df_a(h) + r(h)|)}{|h|} = 0.$$

To prove this, it's tempting to introduce a multiplicative cross-term as follows.

$$\frac{|s(df_a(h) + r(h)|)}{|h|} = \frac{|s(df_a(h) + r(h)|)}{|df_a(h) + r(h)|} \cdot \frac{|df_a(h) + r(h)|}{|h|}$$

But notice that $df_a(h) + r(h) = f(a+h) - f(a)$ by definition of r, so this cross term is problematic for the same reasons as before! If $f(a+h) = f(a)$ for h arbitrary close to 0, we have an infinite sequence of divisions by zeroes that causes problems.

We'll deal with this problem in essentially the same way as before: namely, we extend the definition of

$$\frac{|s(df_a(h) + r(h)|)}{|df_a(h) + r(h)|}$$

so that it is also defined when $f(a+h) = f(a)$.

Second proof of the chain rule, continued. Define $\eta(k) = s(k)/k$. Observe that $\eta(0) = 0$. Moreover, since $s(k) = o(|k|)$ by proposition 2.7, we see that η is continuous at 0. Then

$$\begin{aligned}\frac{|s(df_a(h) + r(h))|}{|h|} &= \frac{|\eta(df_a(h) + r(h))| \cdot |df_a(h) + r(h)|}{|h|} \\ &\leqslant |\eta(df_a(h) + r(h))| \cdot \left(|f'(a)| + \frac{|r(h)|}{|h|}\right).\end{aligned}$$

Observe that $df_a + r$ is continuous at 0 by exercise 2.25 below; moreover, we have just seen that η is continuous at $(df_a + r)(0) = 0$. Thus the composite $\eta \circ (df_a + r)$ is also continuous at 0. Thus

$$\lim_{h \to 0} |\eta(df_a(h) + r(h))| \cdot \left(|f'(a)| + \frac{|r(h)|}{|h|} \right) = 0,$$

where we have used the fact that $|r(h)| = o(|h|)$ to see that the parenthetical expression has a finite limit (namely, $|f'(a)|$). So, by the squeeze theorem, we can therefore conclude that $|s(df_a(h) + r(h))| = o(|h|)$. □

Exercise 2.25. Check that the function r is continuous at 0.

Notice that in the first proof, the trick to overcoming the technicality was defining the function σ at $k = 0$. In the second proof, the trick to overcoming the same technicality was defining η at $k = 0$. These tricks are really the same trick, as shown by the following.

Exercise 2.26. Prove that the function η from the second proof and the function σ from the first proof are related by the equation

$$\sigma(k) = g'(f(a)) + \eta(k).$$

Applications of the chain rule

Exercise 2.27 (Derivatives of inverses)**.** Suppose S and T are both subsets of $\mathbb{R}$ and $f : S \to T$ is bijective, with inverse function $f^{-1} : T \to S$. Suppose further that f is differentiable at an interior point $a \in S$, that $f'(a) \neq 0$, and that $b = f(a)$ is an interior point of T. Prove that f^{-1} is differentiable at b and

$$(f^{-1})'(b) = \frac{1}{f'(a)}.$$

Exercise 2.28 (Power rule for fractional exponents)**.** Extend the result of exercise 2.15 to arbitrary rational exponents (ie, to exponents that can be written in the form m/n where m and n are both integers).

2.C.4. Some rules without proofs

Here are some other rules you probably remember from calculus. We won't prove them here, but you should feel free to use them.

(1) If $f(x) = e^x$, then $f'(x) = e^x$.

(2) If $f(x) = \ln(x)$, then $f'(x) = 1/x$.

(3) If $f(x) = \sin(x)$, then $f'(x) = \cos(x)$.

(4) If $f(x) = \cos(x)$, then $f'(x) = -\sin(x)$.

(5) If $f(x) = x^r$ for some *real* number r, then $f'(x) = rx^{r-1}$.

Note that by doing exercise 2.28, you have already proved the most common cases of the power rule (5).

The proofs of these facts are omitted not because the proofs are particularly difficult, but because we would have to go on a long detour to even define the functions involved sufficiently rigorously to prove anything about them. That would distract us from our object of meditation.

Exercise 2.29. (a) Prove that the function

$$f_1(x) = \begin{cases} x\sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is continuous at 0 but is not differentiable at 0. In particular words, $f \in C^0(\mathbb{R}) \setminus C^1(\mathbb{R})$.

(b) Prove that the function

$$f_2(x) = \begin{cases} x^2\sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is differentiable at 0 but the derivative f_2' is not continuous at 0.

2.D. Levels of differentiability

2.D.1. C^n functions

Definition 2.30. Suppose U is an open subset of $\mathbb{R}$ and $f : U \to \mathbb{R}$ is a function. Then f is *differentiable* if f is differentiable at every $a \in U$. If the function $f' : U \to \mathbb{R}$ is also continuous, then f is *continuously differentiable*.

Definition 2.31. Let U be an open subset of $\mathbb{R}$. For any non-negative integer n, we define the set $C^n(U)$ inductively as follows.

- $C^0(U)$ is the set of continuous functions $f : U \to \mathbb{R}$.

- $C^1(U)$ is the set of functions f such that f' exists and is in $C^0(U)$. Concretely, this means that f is continuously differentiable.
- $C^2(U)$ is the set of functions $f : U \to \mathbb{R}$ such that the derivative f' exists and is in $C^1(U)$. Concretely, this means that the the derivative of the derivative $(f')'$, called the *second derivative* of f and written f'' or $f^{(2)}$, exists and is continuous on U.
- In general, $C^{n+1}(U)$ is the set of functions $f : U \to \mathbb{R}$ such that f' exists and is in $C^n(U)$. This means that the $(n+1)$*st derivative* $f^{(n+1)} = (f^{(n)})'$ exists and is continuous on U.

Exercise 2.32. Prove that, for every non-negative integer n, $C^n(U)$ is a subspace of the vector space of all functions $U \to \mathbb{R}$.

Exercise 2.33. Prove that, if $f, g \in C^n(U)$, then the product fg is also in $C^n(U)$.

We have an infinite chain of inclusions

$$\{\text{all functions } U \to \mathbb{R}\} \supseteq C^0(U) \supseteq C^1(U) \supseteq C^2(U) \supseteq C^3(U) \supseteq \cdots .$$

The following exercise shows that all of these inclusions are strict; in other words $C^n(U) \supseteq C^{n+1}(U)$ for all n.

Exercise 2.34. For every non-negative integer n, let

$$f_n(x) = x^n|x|.$$

Prove that f_n is in $C^n(\mathbb{R})$ but not $C^{n+1}(\mathbb{R})$. *Hint.* Prove that $f'_{n+1} = (n+2)f_n$. Then use induction.

2.D.2. Smooth functions

Definition 2.35. A function $f : U \to \mathbb{R}$ is *infinitely differentiable* or *smooth* if it is in $C^n(U)$ for all n. The set of smooth functions is denoted $C^\infty(U)$ or $\mathcal{O}(U)$.

Exercise 2.36. Consider the function $\psi : \mathbb{R} \to \mathbb{R}$ defined as follows.

$$\psi(x) = \begin{cases} e^{1/(x^2-1)} & \text{if } -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

See figure 2.37 for its graph. Prove that ψ is smooth. *Hint.* First prove directly from definitions that ψ is continuously differentiable. Then use induction to prove that it is in $C^n(\mathbb{R})$ for all n.

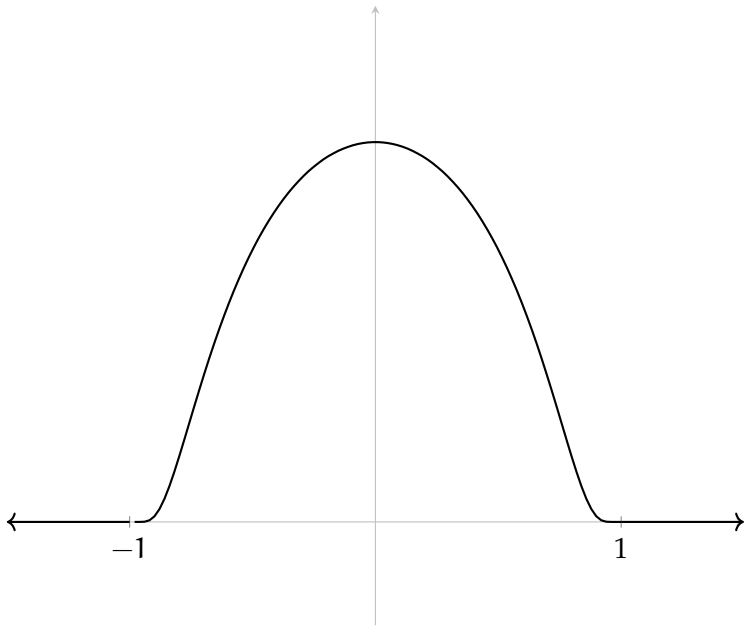

Figure 2.37.: The graph of the function ψ from exercise 2.36.

Exercise 2.38. Let $K = [a, b]$ be a compact interval in $\mathbb{R}$ and let U be an open subset containing K. Prove that there exists a smooth function $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) = 1$ for all $x \in K$ and $f(x) = 0$ for all $x \notin U$. *Hint.* Since U is open, show that there exist $a' < a$ and $b' > b$ in U. Thus, if we let $I = [a', b']$, then $K \subsetneq I \subseteq U$. The characteristic function

$$\chi_I(x) = \begin{cases} 1 & \text{if } x \in I \\ 0 & \text{if } x \notin I \end{cases}$$

is not smooth, since it's not even continuous; obtain f by modifying χ_I on the intervals $[a', a]$ and $[b, b']$ using inspiration from the function ψ of exercise 2.36.

3. Multivariable derivatives

3.A. Definition of the derivative

Throughout, S will denote a subset of $\mathbb{R}^m$.

Definition 3.1. A function $f : S \to \mathbb{R}^n$ is *differentiable at* an interior point $a \in S$ if there exists a linear map $\ell : \mathbb{R}^m \to \mathbb{R}^n$ such that

$$|f(a+h) - f(a) - \ell(h)| = o(|h|) \text{ as } h \to 0.$$

It turns out that there can exist at most one linear map that has the above property.

Lemma 3.2. *Suppose $f : S \to \mathbb{R}^m$ is differentiable at an interior point $a \in S$. If ℓ and ℓ' are both linear maps $\mathbb{R}^m \to \mathbb{R}^n$ such that*

$$|f(a+h) - f(a) - \ell(h)| = o(|h|) \textit{ and } |f(a+h) - f(a) - \ell'(h)| = o(|h|) \textit{ as } h \to 0,$$

then $\ell = \ell'$.

Proof. Let $\phi = \ell' - \ell$. Then ϕ is also a linear map $\mathbb{R}^m \to \mathbb{R}^n$. Moreover, observe that

$$\begin{aligned} |\phi(h)| &= |\ell'(h) - \ell(h)| \\ &= |(f(a+h) - f(a) - \ell(h)) - (f(a+h) - f(a) - \ell'(h)| \\ &\leqslant |f(a+h) - f(a) - \ell(h)| + |f(a+h) - f(a) - \ell'(h)|. \end{aligned}$$

Since both $|f(a+h) - f(a) - \ell(h)|$ and $|f(a+h) - f(a) - \ell'(h)|$ are $o(|h|)$, exercises 1.14 and 1.15 imply that $|\phi(h)|$ is also $o(|h|)$. Then exercise 1.23 implies that $\phi = 0$. □

Definition 3.3. Suppose $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$. Then the *derivative* of f at a, denoted df_a, is the unique linear map $\mathbb{R}^m \to \mathbb{R}^n$ satisfying

$$|f(a+h) - f(a) - df_a(h)| = o(|h|) \text{ as } h \to 0.$$

This definition is very difficult to compute examples from. Virtually the only concrete example that we can calculate without much effort directly from the definition is the following.

Exercise 3.4. Suppose $\ell : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map. Prove that ℓ is differentiable at every $a \in \mathbb{R}^m$ and $d\ell_a = \ell$.

So we are now going to embark upon a journey to find a more concrete description of the derivative; ultimately, we will calculate an explicit matrix representation for the derivative whose entries will be single-variable derivatives. But our journey will be arduous and abstract. So, before we embark, let us explore a concrete example.

3.B. Detailed example

3.C. Rules of differentiation

3.C.1. Sum and scalar multiples rule

Exercise 3.5 (Sum rule)**.** Prove that, if $f, g : S \to \mathbb{R}^n$ are both differentiable at an interior point $a \in S$, then $f + g$ is also differentiable at a and

$$d(f+g)_a = df_a + dg_a.$$

Exercise 3.6 (Scalar multiples rule)**.** Prove that, if c is a constant and $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$, then cf is also differentiable at a and

$$d(cf)_a = c \cdot df_a.$$

3.C.2. Chain rule

The proof of the following is essentially the same as the second proof of the single variable chain rule 2.16. Since this proof is basically a repeat, some details are omitted; you are asked to fill in the details in exercise 3.9 below.

Proposition 3.7 (Chain rule)**.** *Suppose that S and T are subsets of $\mathbb{R}^m$ and $\mathbb{R}^n$, respectively, that $f : S \to T$ is differentiable at an interior point a, that $f(a)$ is an interior point of T, and that $g : T \to \mathbb{R}^p$ is differentiable at $f(a)$. Then the composite $g \circ f : S \to \mathbb{R}^p$ is also differentiable at a, and*

$$d(g \circ f)_a = dg_{f(a)} \circ df_a.$$

Proof. Define the "error in approximation" functions

$$\begin{aligned} r(h) &= f(a+h) - f(a) - df_a(h) \\ s(k) &= g(f(a)+k) - g(f(a)) - dg_{f(a)}(k) \end{aligned}$$

and then observe that

$$g(f(a+h)) - g(f(a)) - dg_{f(a)}(df_a(h)) = dg_{f(a)}(r(h)) + s(df_a(h) + r(h)). \tag{3.8}$$

This means that

$$|g(f(a+h)) - g(f(a)) - dg_{f(a)}(df_a(h))| \leqslant |dg_{f(a)}(r(h))| + |s(df_a(h) + r(h))|.$$

Since $dg_{f(a)}$ is linear and $|r(h)| = o(|h|)$, exercise 1.21 guarantees that $|dg_{f(a)}(r(h))| = o(|h|)$ also. Thus it is sufficient to prove that $|s(df_a(h) + r(h))| = o(|h|)$. To do this, define $\eta(k) = |s(k)|/|k|$ and then notice that

$$\begin{aligned} \frac{|s(df_a(h) + r(h))|}{|h|} &= \eta(df_a(h) + r(h)) \cdot \frac{|df_a(h) + r(h)|}{|h|} \\ &\leqslant \eta(df_a(h) + r(h)) \left(\|df_a\| + \frac{|r(h)|}{|h|} \right). \end{aligned}$$

We now take the limit as $h \to 0$. Using the facts that $\eta \circ (df_a + r)$ is continuous at 0 and that $|r(h)| = o(|h|)$, we obtain the result. □

Exercise 3.9. (a) Prove equation (3.8).

(b) If you haven't done it already, do exercise 1.21.

(c) Prove that $\eta \circ (df_a + r)$ is continuous at 0.

3.C.3. Differentiability by components

In this section, we'll show that it's "enough" to consider the case $n = 1$.

Definition 3.10. For $j = 1, \ldots, n$, let $\pi_j : \mathbb{R}^n \to \mathbb{R}$ denote the jth *projection map*

$$\pi_j(x_1, \ldots, x_n) = x_j.$$

Then, for any function $f : S \to \mathbb{R}^n$, we define the jth *component function* $f_j : S \to \mathbb{R}$ to be the composite $\pi_j \circ f$.

Proposition 3.11. *A function $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$ if and only if the component function $f_j : S \to \mathbb{R}$ is differentiable at a for all $j = 1, \dots n$. Moreover,*

$$df_a(h) = \begin{bmatrix} df_{1,a}(h) \\ \vdots \\ df_{n,a}(h) \end{bmatrix}. \tag{3.12}$$

Proof. If f is differentiable at a, then the composite $f_j = \pi_j \circ f$ is also differentiable at a by the chain rule 3.7. Moreover, since π_j is linear, we know that $d\pi_{j,f(a)} = \pi_j$ by exercise 3.4. Thus, by the chain rule,

$$df_{j,a}(h) = (d\pi_{j,f(a)} \circ df_a)(h) = \pi_j(df_a(h))$$

for all j, which is precisely equation (3.12). For the converse, we turn equation (3.12) on its head. Suppose f_j is differentiable at a for all j, and let $\ell : \mathbb{R}^m \to \mathbb{R}^n$ be the linear map

$$\ell(h) = \begin{bmatrix} df_{1,a}(h) \\ \vdots \\ df_{n,a}(h) \end{bmatrix}.$$

We now want to show that $|f(a+h) - f(a) - \ell(h)| = o(|h|)$ as $h \to 0$. □

3.D. Partial derivatives

Definition 3.13. Suppose $f : S \to \mathbb{R}$ is a function and $a \in S$ is an interior point.

Proposition 3.14. *If $f : S \to \mathbb{R}$ is differentiable at a, then $\partial_i f(a)$ exists for all $i = 1, \dots m$.*

3.E. Jacobian matrix

Definition 3.15. Suppose $f : S \to \mathbb{R}^n$ is differentiable at an interior point $a \in S$. The *Jacobian matrix* at a, denoted $J(a)$, is the $n \times m$ matrix whose (j, i)-entry is $\partial_i f_j(a)$. In other words,

$$J(a) = \begin{bmatrix} \partial_1 f_1(a) & \partial_1 f_2(a) & \cdots & \partial_1 f_n(a) \\ \partial_2 f_1(a) & \partial_2 f_2(a) & \cdots & \partial_2 f_n(a) \\ \vdots & \vdots & \ddots & \vdots \\ \partial_m f_1(a) & \partial_m f_2(a) & \cdots & \partial_m f_n(a) \end{bmatrix}$$

It follows from proposition 3.14 that all of these partial derivatives exist.

Theorem 3.16. *Suppose* $f : S \to \mathbb{R}^m$ *is differentiable at an interior point* $a \in S$*. Then*

$$[df_a] = J(a).$$

3.F. Inverse function theorem

3.G. Levels of differentiability

Throughout, U denotes an open subset of $\mathbb{R}^m$.

Definition 3.17. If a function $f : U \to \mathbb{R}^n$ is differentiable at every $a \in U$, then f is *differentiable on* U. Then the function $U \to \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ given by $a \mapsto df_a$ is denoted df.

Definition 3.18. If $f : U \to \mathbb{R}^n$ is a differentiable function and $df : U \to \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)$ is continuous, then f is *continuously differentiable*.

Lemma 3.19. *A function* $f : U \to \mathbb{R}^n$ *is continuously differentiable if and only if* $\partial_i f_j : U \to \mathbb{R}$ *is continuously differentiable.*

4. Manifolds

5. Tangent spaces

Bibliography